3 **(a)** **(i)**

$$F = \frac{GMm}{(R+r)^2}$$ B1

(ii) M has a slower speed due to its smaller orbital radius and hence has a smaller acceleration. By Newton's second law, for the same gravitational force between M and m, M must be of a bigger mass. B1 B1

or

Alternatively, since both masses experience the same gravitational force (N3L) → same centripetal force → since r is smaller than R, M has to be larger than m. B1 B1

*Can use centre of mass to explain?

(iii) For star P,

$$\frac{GMm}{(R+r)^2} = Mr\omega^2$$

Centripetal acceleration for star P = $\frac{Gm}{(R+r)^2} = r\omega^2$ --- (1) A1

For star Q,

$$\frac{GMm}{(R+r)^2} = mR\omega^2$$

Centripetal acceleration for star Q = $\frac{GM}{(R+r)^2} = R\omega^2$ --- (2) A1

(iv) Gravitational force on each star provides for their individual centripetal force. C1

Adding (1) and (2),

$$\frac{G(M+m)}{(R+r)^2} = (R+r)\omega^2 = (R+r)\left(\frac{2\pi}{T}\right)^2$$ A1

$$T = \sqrt{\frac{4\pi^2(R+r)^3}{G(M+m)}}$$

(b) Let m_r be mass of rocket.

loss in E_k of rocket = gain in gpe of rocket

$0.5m_r v^2 - 0 = 0 - [m_r\,(GM/r + Gm/R)]$

$0.5m_r v^2 = [Gm_r\,[(4.50 \times 10^{23}/4.0 \times 10^8) + (1.50 \times 10^{23})/(1.2 \times 10^9)]]$ C1

$0.5m_r v^2 = Gm_r\,(1.25 \times 10^{15})$ A1

$v = 408\ \text{m s}^{-1}$